Introduction to Integrated Ethics Labs

Integrated Ethics Team

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The Integrated Ethics Curriculum for the Computer and Data Sciences is an on-going project with the goal of providing professor and student-friendly ethics labs that can be integrated into computer and data science core courses. We believe, as have others before us, that ethics have become an increasingly important part of the undergraduate education of students in these disciplines. Furthermore, we believe that ethics studies should be progressive, and included in all four years of the student experience.

Curriculum Structure

Year 1: Laying a foundation

While a particular lab can be used in isolation, our experience shows that laying the groundwork with the first year curriculum prepares the students to gain more from subsequent course-correlated labs. If you do choose to use a lab in isolation, we recommend that you have the students complete the A First Look at Ethics in Computer and Data Sciences lab as homework prior to presenting the isolated lab.

The first year of the curriculum introduces a toolkit of ethical frameworks for use in evaluating approaches to ethical dilemmas. The use of these framework tools is encouraged throughout the curriculum as a way of pushing students to reason through the big picture and potential solutions rather than simply voicing an impulsive opinion. The most useful framework for a given situation can then be used to support and articulate a more thoughtful opinion.

Four frameworks are presented first in year one and then used and reinforced in the subsequent years. The first three frameworks are frequently used in ethics textbooks, but we see the use of analogies as complementary. The virtue ethics framework encourages the student to answer the questions “What approach would a virtuous person take?” and “Which virtues would this decision uphold?” The consequentialism framework asks the student to consider who this decision would affect and to try to determine if the benefits of the outcome to some outweigh the costs to others. In other words, does the chosen approach bring the most “utility” or happiness to the most people? The deontological framework informs a decision based on a set of rules such as state laws or religious commandments or a company code of ethics.

Sometimes, before any of these frameworks can be used, it is helpful to just understand the situation better. The computer and data science fields are new enough and dynamic enough that there may be limited or no precedence for dealing with a particular situation. This is where an analogy can be helpful. An analogy is saying something is like something else for the purpose of clarification. If a concept is better understood in an analogous situation than in the dilemma in question, the analogy might help in understanding the less-familiar predicament.

Years 2 - 4: Applications

The labs in years two through four are more applied. They are designed to fit with the content of a particular course. For example, an ethics lab to be integrated into a Data Analysis course might deal with the potential for unethical choices in data cleaning, while a lab to be integrated into a Software Engineering course might deal with the addictive aspects of certain software products.

Recommendations for using the labs

We recommend that no faculty member be "forced" to present one of the labs. It is important for the students to see the labs as a significant and integral part of the course. A faculty member who presents the material under duress can influence students negatively.

Recognize that you are presenting material that is interdisciplinary in nature. While you are the content (computer, statistics, data science) expert, you are not an expert in ethics. You will feel awkward at times in presenting the ethics side of the lab. However, this makes you an excellent role model for students who are also not ethics experts but who will need to address ethical issues in their future occupations. Comfort will be gained as you present a lab a second or third time and are able to adapt it to your personal style.

The ethics labs employ a variety of activities, and feedback shows that students find them to be quite engaging. It is often the case that students who rarely speak up in class will be quite vocal with this material. Therefore, while each lab could be completed in about 30 minutes, be prepared to be flexible regarding time.

Each lab provides to the professor the following:

- Ethics and content knowledge prerequisites
- Learning objectives
- An overview of the lab
- A step-by-step guide for administering the lab
- Student handouts, if applicable
- Assessment ideas

While the labs are designed to be somewhat “off-the-shelf”, the best results occur when the professor prepares in the following ways:

- Read the entire lab several times to familiarize yourself with the material.
- Print or post online any student handouts.
- Adapt the lab for the size and personality of the course.
- If a lab is used without any previous introduction to ethics, we recommend that you have the students complete the A First Look at Ethics in Computer and Data Sciences lab as homework prior to presenting the isolated lab.

Bibliography